



Jashore University of Science and Technology
Department of Computer Science and Engineering

Semester Final Examination-2025
3rd Year 1st Semester, Session: 2021-2022

Course Code: CSE 3105

Time: 3.00 hours

Course Title: Microprocessors and Embedded System

Marks: 72

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

1.
 - a) Define Microprocessor. Analyze different parts of the microprocessor. 03
 - b) Explain the role of the ALU and Control Unit in the 8086 architecture. 03
 - c) Draw the block diagram of the 8086 microprocessor and label its components. 03
 - d) What are the differences between 8086 and Pentium microprocessor? 03
2.
 - a) Define the term segment register and explain four segment registers in 8086 microprocessor. 2+3
 - b) Describe the function of the Bus Interface Unit (BIU) and Execution Unit (EU). 04
 - c) Briefly explain the general-purpose registers of 8086 microprocessor. 03
3.
 - a) Write the syntax of the AND, TEST, and XCHG instructions in 8086. 03
 - b) Differentiate between ADD and ADC instructions. 03
 - c) Describe the working of the CALL and RET instructions in the 8086 microprocessor. How do they affect the stack and program execution flow? 06
4.
 - a) Define cache memory and mention its role in computer operations. 02
 - b) Suppose two memories of size 1KB and 4KB need to be connected to a microprocessor having 16 address lines. Design a memory interfacing circuit to uniquely communicate with each memory, briefly explain its function, and find address ranges. 04
 - c) Design an I/O interfacing circuit using a 3 to 8 decoder to handle an output at port address 05H and an input device at port address 07H. Briefly explain its operation. 03
 - d) Define NMI, INTR, and INTA. 03
5.
 - a) What is TEST instruction and how it works, explain with an example. 03
 - b) Check the following code below. If there is any mistake, mention the errors and rewrite the program to solve the mistakes. Analyze the output of the program. 05

```
1. .MODEL SMALL
2. .STACK 100H
3. .DATA
4. MAIN PROC
5.     MOV AH, 2
6.     CMP AX, 0
7.     JL NEGATIVE
8.     JE ZERO
9.     JG POSITIVE
10.    NEGATIVE:
11.        MOV BX, -1
12.    ZERO:
13.        MOV BX, 0
14.        JMP END_CASE
15.    POSITIVE:
16.        MOV BX, 1
17.
18.    MOV AH, 4CH
19.    INT 21H
20.    MAIN ENDP
21. END MAIN
```

- c) Describes how load string and scan string works. Explain with figures.

04

✓ 6. a) Write an 8086 assembly program to add two 16-bit numbers stored in memory locations NUM1 and NUM2, and store the result at location SUM. Assume the data segment is initialized. 04

b) Explain the function of the following instructions: 04

i) MOV AX, BX ii) MUL BL iii) SUB AL, 05H iv) JZ NEXT

c) Write an 8086 assembly program to read a character from the keyboard and display it back on the screen using DOS interrupt INT 21H. 04

✓ 7. a) Write an 8086 assembly program to multiply two 8-bit numbers stored in NUM1 and NUM2 and store the 16-bit result in RESULT. 04

b) Consider the following code snippet: 04

```
MOV AX, 0000H
MOV CX, 0005H
MOV BX, 0002H
LOOP1: ADD AX, BX
INC BX
LOOP LOOP1
```

Determine:

- i) The final value of AX in hexadecimal.
- ii) How many times the ADD instruction executes.

e) Write an 8086 assembly program to display the message "HELLO" on the screen using DOS interrupt INT 21H. 04

✓ 8. a) Show the comparison between 8086 and 8088 microprocessors. 04

b) Explain the basic features of Intel 8087 Co-processor that work alongside the Intel 8086. 04

c) Describe the Bus Cycle Status Lines $\overline{S_0} - \overline{S_2}$ and Queue Status Lines $QS_0 - QS_1$. 04

MRW P